Mathematics-1 for Data Science

Week 2 Notes

Course: Mathematics–1

Notes created by: Vishal (Instructor)

IIT Madras BS Degree Programme

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Lecture 1 — Elements of Coordinate Geometry

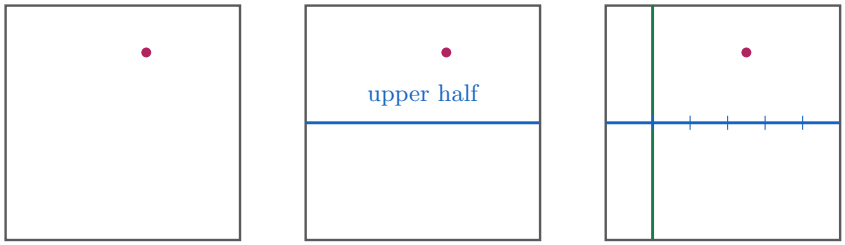
The rectangular coordinate system: axes, origin, ordered pairs and the four quadrants

What this lecture covers

Today we look at some elements of **coordinate geometry**, identifying them as axes, points and lines. We have already seen in basic geometry what points, lines and planes are; here we study the *algebraic* properties of these geometric objects. In that context we first revise the **rectangular coordinate system** — why it matters and how to use it.

1 Describing the Position of a Point

Given a point on a plane, we want to describe its location. As things stand, we cannot say much more than "this point lies slightly towards the top right of the plane". Let us improve the description step by step.



1. a bare plane **2.** a horizontal line **3.** a number scale

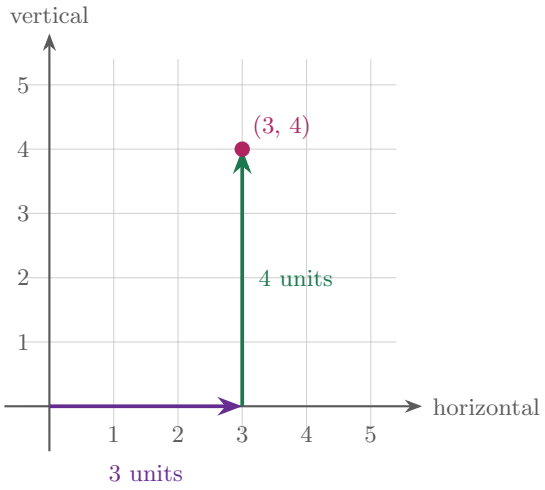
- Introducing a **horizontal line** lets us say the point is in the upper half of the plane — a slightly better description.
- Associating a **real number system** with that line lets us say the point lies between 0 and 5.
- Adding a **perpendicular line**, also carrying a real number system, narrows the point down to a square enclosed by 5 on the vertical line and 5 on the horizontal line.
- Finally, drawing in the **grid lines** pins down the exact location.

Reading off the exact location

Travelling 3 units in the horizontal direction and 4 units in the vertical direction brings us to the point. So we may name the point

$$(3,\ 4)$$

which is a precise description of it.



What we have built is a **reference system** through which we can specify the location of a point precisely.

2 The Rectangular Coordinate System

Travelling 3 units horizontally and 4 units vertically means we are specifying a coordinate in the horizontal direction and a coordinate in the vertical direction. These directions have names.

Axes, origin and coordinates

- The horizontal line is the X **axis**; it allows movement from left to right.
- The vertical line is the Y **axis**; it allows movement up and down.
- Their point of intersection is called the **origin**.
- Any point of the plane is denoted by an **ordered pair**

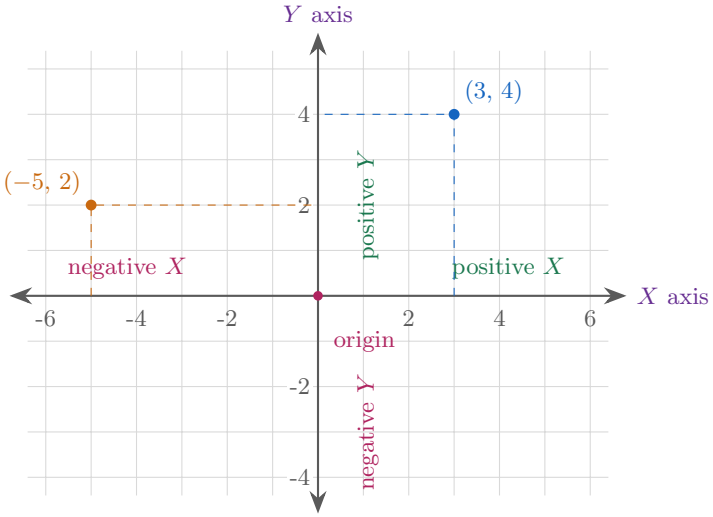
$$(x,\ y)$$

where x is the distance travelled horizontally — the x **coordinate** — and y is the distance travelled vertically — the y **coordinate**. Hence the name *coordinates*.

The vertical line cuts the horizontal line into two parts: the **positive part of the X axis** and the **negative part of the X axis**. Similarly the horizontal line cuts the vertical line into a positive part of the Y axis above and a negative part below. This whole structure is called the **coordinate plane**.

Why "rectangular"?

The X axis and the Y axis meet each other at an angle of 90° — the two lines are perpendicular. The Latin *recta* means *right*, so **rectangular** here means a 90° coordinate system: a rectangular coordinate system.



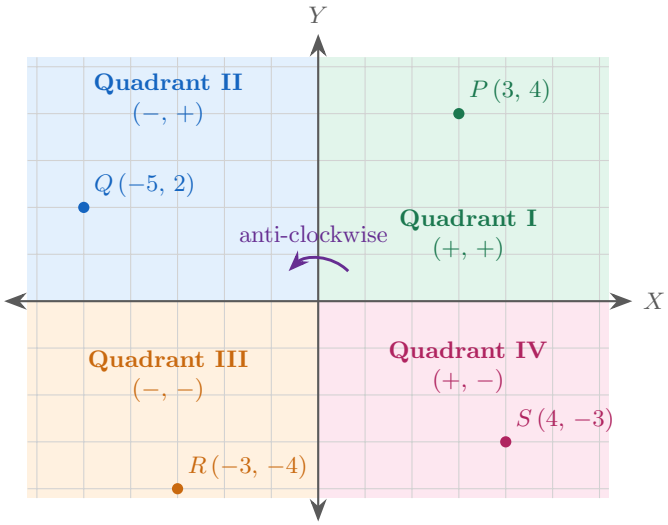
Locating two points

The point $(3, 4)$**.** We travel 3 units in the horizontal direction and 4 units in the vertical direction.
The point $(-5, 2)$**.** Here the x coordinate is negative, so we go to the *left* of the vertical line, travelling a distance of 5 units. The y coordinate is positive, so on the upper half we travel 2 units up. That gives the point $(-5, 2)$.

This is how we can uniquely describe points using the coordinate plane.

3 The Four Quadrants

Looking at the two points above, we see that the coordinate axes divide the coordinate plane into four parts. By mathematical convention we move round them in the **anti-clockwise** direction.



The four sample points

$P\,(3,\ 4)$	lies in Quadrant I	both coordinates positive
$Q\,(-5,\ 2)$	lies in Quadrant II	x negative, y positive
$R\,(-3,\ -4)$	lies in Quadrant III	both coordinates negative
$S\,(4,\ -3)$	lies in Quadrant IV	x positive, y negative

An easy way to remember

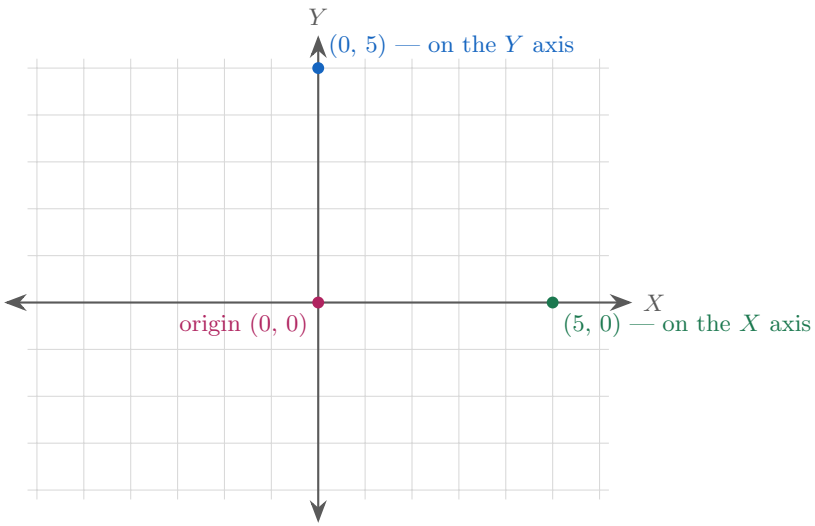
The **odd** quadrants — I and III — have x and y coordinates of the **same sign**: both positive in Quadrant I, both negative in Quadrant III.
The **even** quadrants — II and IV — have coordinates of **opposite signs**: one coordinate positive and the other negative.

4 Points on the Axes

A question may now arise. Take the point $(5, 0)$ — in which quadrant does it lie?

Points on the axes belong to no quadrant

The point $(5, 0)$ does **not** lie in any quadrant: it lies *on the X axis*.
Similarly $(0, 5)$ lies in no quadrant either — it lies *on the Y axis*.



So the coordinate plane is subdivided into **four quadrants** and **two axes**, with one remaining point — the point of intersection, the **origin**.

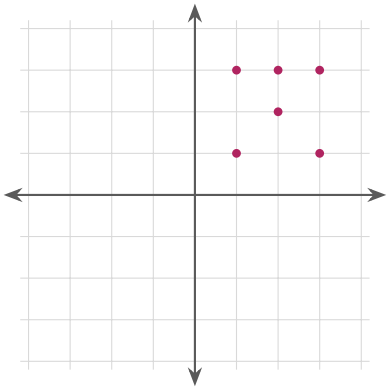
Region	x coordinate	y coordinate
Quadrant I	positive	positive
Quadrant II	negative	positive
Quadrant III	negative	negative
Quadrant IV	positive	negative
X axis	positive or negative	0
Y axis	0	positive or negative
Origin	0	0

5 Why the Quadrant System Helps

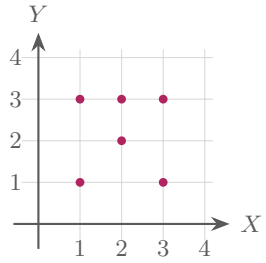
Sometimes we are given several points to plot. Looking at them closely, we need not divide the system in an equally spaced manner across the whole plane.

Scaling the picture to the data

If many of the points lie in Quadrant I, we can **scale** the picture — bringing the origin to the bottom left corner — and focus on Quadrant I alone.



all four quadrants drawn



Quadrant I alone, rescaled

So with a good understanding of quadrants we can graph points, and later functions, far better — and that is why the coordinate system is important.

Lecture 2 — The Distance Formula

Distance of a point from the origin, and the distance between any two points

What this lecture covers

After the coordinate system, let us take up one classical problem. If a point is located at, say, $(3, 4)$ and somebody asks what its distance from the origin is, how do we answer? We build the answer out of Pythagoras' theorem, generalise it to an arbitrary point (x, y), and then extend it further to the distance between *any* two points of the coordinate plane.

1 Distance of a Point from the Origin

Our goal is to find the distance of the point $P\,(3, 4)$ from the origin. That reduces to finding the length of the line segment joining O and P. Is there a classical tool that helps?

The easy cases first

Suppose the point lay on one of the axes.

Points on the axes

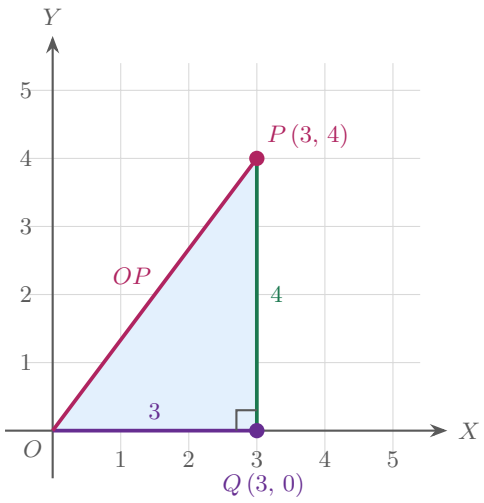
If the point were $(3, 0)$, we would know its distance at once — we simply count the units travelled in the horizontal direction, so the distance is 3.
If the point lay on the Y axis, we would again know the answer — we count the number of units we must travel vertically to reach it.

So for a point on the X axis or the Y axis we already know how to find the distance from the origin. The question is what to do when the point lies anywhere in the coordinate plane.

Bringing in a right-angled triangle

The trick is to relate the position of the point to the coordinate axes. Take the **image** of $(3, 4)$ on the X axis: the easiest way is to **drop a perpendicular** onto the X axis, which meets it at the point $Q\,(3, 0)$.

Once that is done, the X axis and the vertical line together with OP form a **right-angled triangle**.



Pythagoras' theorem

In a right-angled triangle, the length of the hypotenuse is the square root of the sum of the squares of the lengths of the adjacent sides.

Applying it to the triangle OQP:

$$OP^2 = OQ^2 + QP^2$$

- OQ is a line segment lying along the X axis, and its length is 3 units.
- QP is parallel to the Y axis — as good as a projection onto the Y axis — and its length is 4 units.

Working it out

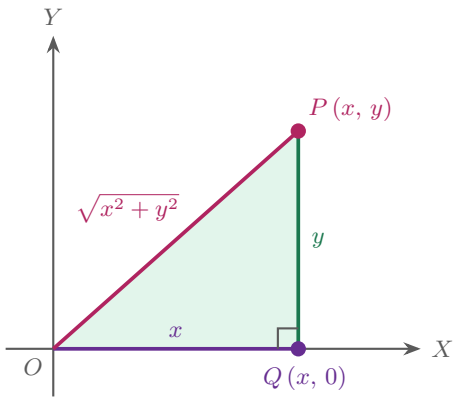
$$OP = \sqrt{OQ^2 + QP^2} = \sqrt{3^2 + 4^2} = \sqrt{9 + 16} = \sqrt{25} = 5$$

taking the positive square root.

Generalising

Is there anything special about the point $(3, 4)$? No — nothing at all. We could have started with a point $P\,(x, y)$ and taken its image on the X axis, which is $Q\,(x, 0)$. Then the length of OQ is x, and the length of QP runs from 0 to y, so it is y units.

$$OP = \sqrt{x^2 + y^2}$$



2 Distance Between Any Two Points

Let us now generalise the problem: find the distance between any two points

$$P\,(x_1,\ y_1) \qquad \text{and} \qquad R\,(x_2,\ y_2).$$

To make things concrete, take

$$(x_1,\ y_1) = (5,\ 6) \qquad \text{and} \qquad (x_2,\ y_2) = (-1,\ 2).$$

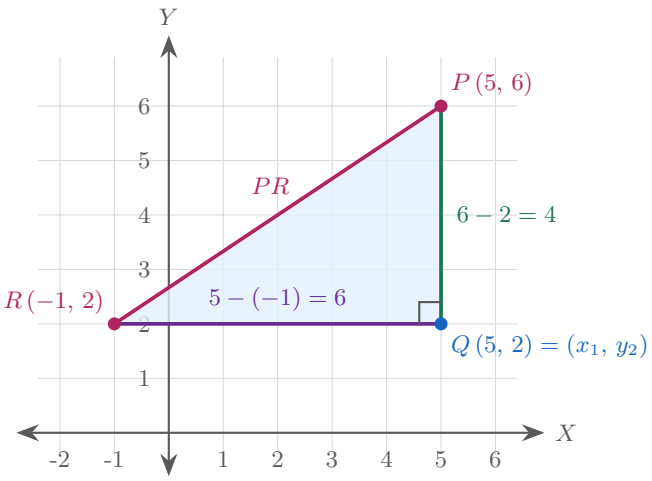
One easy way to find the distance is again to construct a right-angled triangle. But now neither point lies on an axis — $(-1, 2)$ is on no axis at all — so we cannot simply "drop a perpendicular to the X axis".

The careful construction, and the shortcut

We could drop a perpendicular from P to the X axis, meeting it at $(5, 0)$, and then drop a perpendicular from $R(-1, 2)$ onto *that* line, meeting it where the y coordinate is 2 and the x coordinate is 5 — namely at $(5, 2)$ — giving a right-angled triangle.
Skipping those steps, we can say directly: **construct a right-angled triangle with the right angle at**

$$Q = (x_1,\ y_2)$$

which in this example is the point $(5, 2)$.
This phrasing is better because the second point may well lie in the third quadrant, and then dropping a perpendicular to the X axis would not help — we would have to extend it beyond the axis. Stating the right angle at (x_1, y_2) works wherever the point lies.



Applying Pythagoras again

With the right-angled triangle in place, the same theorem comes into play:

$$PR^2 = QR^2 + PQ^2$$

Can we compute QR and PQ? Yes — because QR is parallel to the X axis and PQ is parallel to the Y axis, so computing them is as good as computing a length along the respective axis. Each is simply the **change in the coordinate**.

Which way to subtract

When computing a length parallel to the X axis, go from **left to right** — subtract taking the higher value first. When computing a length in the vertical direction, go from **bottom to top** — subtract the lowest y value from the highest.

The two legs

$$QR = 5 - (-1) = 6 \text{ units} \qquad PQ = 6 - 2 = 4 \text{ units}$$

The distance formula

In this case we are squaring the differences, so it does not actually matter which coordinate we take first: even a negative value becomes positive on squaring. So for general coordinates (x_1, y_1) and

(x_2, y_2) we may write $x_2 - x_1$ and $y_2 - y_1$, regardless of which is bigger, and take the positive square root.

$$PR = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Back to the example

$$PR = \sqrt{6^2 + 4^2} = \sqrt{36 + 16} = \sqrt{52} = 2\sqrt{13}$$

The distance formula

For any two points of the coordinate plane,

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

and putting $(x_1, y_1) = (0, 0)$ recovers the first case, the distance of (x, y) from the origin:

$$d = \sqrt{x^2 + y^2}$$

Lecture 3 — The Section Formula

A point dividing a line segment in a given ratio, derived from similar triangles

What this lecture covers

We have handled two points; now let us take *three*. Suppose the three points lie on a line, and the point P cuts the line segment AB in the ratio $m : n$. Our goal is to find the coordinates of P — and that gives us the **section formula**.

1 Setting Up the Problem

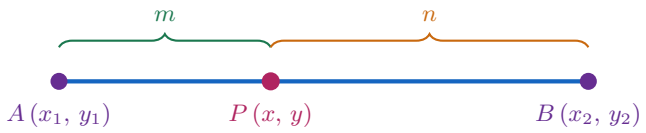
There is a line segment AB, and a point P cuts it in the ratio $m : n$. How do we find the coordinates of P? Let us bring in our coordinate system.

The set-up

$$A = (x_1,\ y_1), \qquad B = (x_2,\ y_2), \qquad P = (x,\ y)$$

where x and y are unknown, and

$$\frac{AP}{PB} = \frac{m}{n} \quad \text{is given.}$$



2 Building Two Right-Angled Triangles

To understand the situation in terms of the coordinate axes, we construct two triangles using the same logic as in the distance formula.

- From A and P: construct a right-angled triangle whose third vertex Q has the x coordinate of P and the y coordinate of A, that is

$$Q = (x,\ y_1).$$

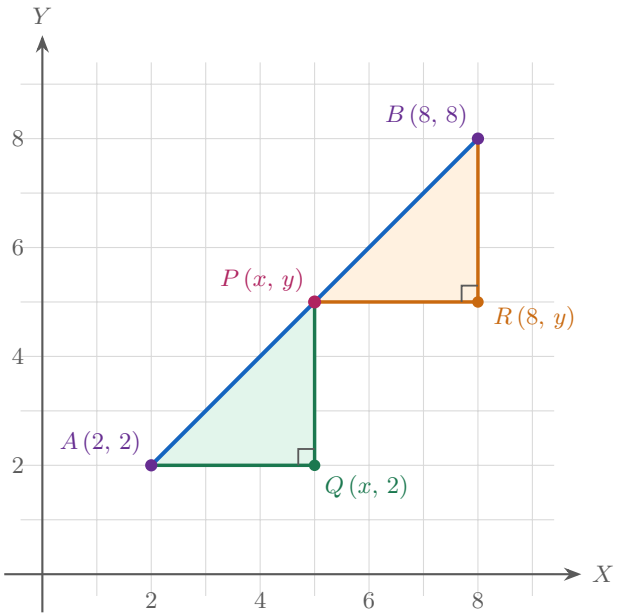
- From P and B: drop a perpendicular from B and construct a right-angled triangle whose third vertex R has the y coordinate of P and the x coordinate of B, that is

$$R = (x_2,\ y).$$

For simplicity let us plot the points with concrete references: take

$$A = (2,\ 2), \qquad B = (8,\ 8),$$

so that $Q = (x,\ 2)$ and $R = (8,\ y)$.



3 Why the Triangles Are Similar

The angle argument

- The line AQ is parallel to the X axis, and the line PR is parallel to the X axis as well. So these two are **parallel lines**, and the segment AB is a **transversal** passing through them. Hence the angle at A and the angle at P are equal.
- The angles at Q and at R are both **right angles**.
- Since the angles of a triangle sum to $180°$, the remaining angle at B must equal the remaining angle at P.

Therefore, by the angle test,

$$\triangle AQP \sim \triangle PRB$$

which means their corresponding sides are in the same ratio.

$$\frac{AP}{PB} = \frac{AQ}{PR} = \frac{PQ}{RB}$$

Here AP and PB are the hypotenuses of the two right-angled triangles.

4 Computing the Sides

We already know that $AP : PB = m : n$. Can we calculate the lengths of the other sides? Yes — because AQ and PR are parallel to the X axis, and PQ and BR are parallel to the Y axis, so each is just a difference of coordinates, taking the higher value first.

The four sides

Horizontal sides (subtract along the X direction):

$$AQ = x - 2 = x - x_1 \qquad PR = 8 - x = x_2 - x$$

Vertical sides (subtract along the Y direction):

$$PQ = y - 2 = y - y_1 \qquad BR = 8 - y = y_2 - y$$

Putting everything together:

$$\frac{m}{n} = \frac{x - x_1}{x_2 - x} = \frac{y - y_1}{y_2 - y}$$

5 Deriving the Section Formula

Take one equality at a time. First the x coordinates:

$$\frac{m}{n} = \frac{x - x_1}{x_2 - x}$$

Cross-multiplying and rearranging:

$$m\,(x_2 - x) = n\,(x - x_1) \qquad \Longrightarrow \qquad mx_2 - mx = nx - nx_1$$

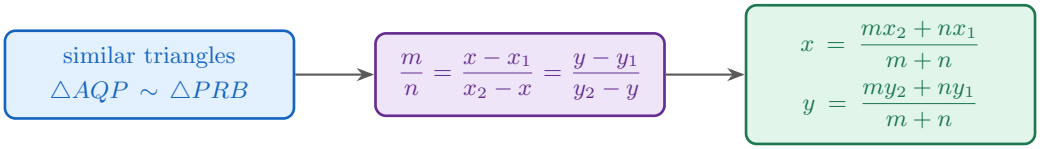
$$mx_2 + nx_1 = (m + n)\,x$$

Doing the same with the ratio of the y coordinates gives the companion result.

The section formula

If P divides the line segment joining $A\,(x_1, y_1)$ and $B\,(x_2, y_2)$ in the ratio $m : n$, then

$$x = \frac{m\,x_2 + n\,x_1}{m + n} \qquad\qquad y = \frac{m\,y_2 + n\,y_1}{m + n}$$



The reverse question

Another interesting question: suppose we know the coordinates x and y — can we find the ratio in which the point divides the segment? Yes. Since the coordinates are known, we simply use the same formula to solve for the ratio. This becomes clearer with practice on more problems.

Lecture 4 — Area of a Triangle

Three non-collinear points, three trapeziums, and the coordinate formula for area

What this lecture covers

After the section formula, let us understand three points when they are *not* on one line — that is, when they form a triangle. Given three points known to be non-collinear, how do we find the area of the triangle using the coordinate system? The answer comes from dropping perpendiculars to the X axis and reading off trapeziums.

1 Setting Up

Let the triangle be ABC with coordinates

$$A = (x_1,\ y_1), \qquad B = (x_2,\ y_2), \qquad C = (x_3,\ y_3).$$

With these coordinates we can plot it. How do we now find its area?

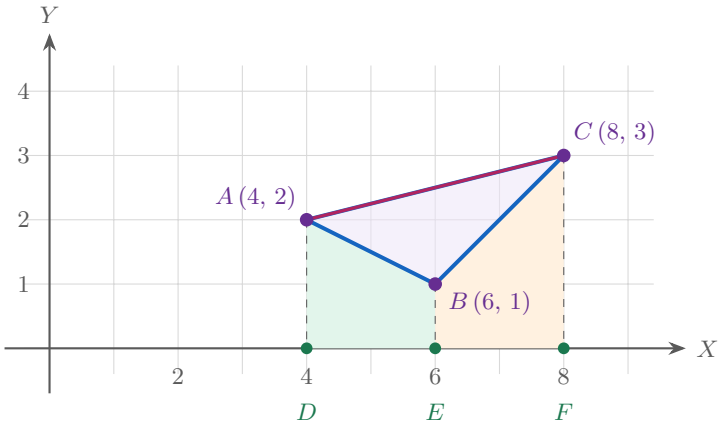
Everything we have discussed so far relied on **dropping a perpendicular to the X axis** and finding the area of the geometric object formed — in earlier cases just a triangle. Following the same idea, we drop perpendiculars from all three vertices to the X axis, meeting it at

$$D = (x_1,\ 0), \qquad E = (x_2,\ 0), \qquad F = (x_3,\ 0).$$

This generates three **trapeziums**:

- $ADFC$ — the biggest one, encompassing everything;
- $ADEB$ — on the left;
- $BEFC$ — on the right.

Our triangle is trapped in between them.



The strategy

To find the area of triangle ABC, take the biggest trapezium $ADFC$ and remove the two smaller ones:

$$\text{ar}(\triangle ABC) = \text{ar}(ADFC) - \text{ar}(ADEB) - \text{ar}(BEFC)$$

Whatever is left is the area of the triangle.

About this picture

This decomposition is drawn for one particular arrangement: all three vertices above the X axis, with $x_1 < x_2 < x_3$ and the middle vertex B lying *below* the side AC. Other arrangements need a slightly different combination of trapeziums, but every one of them leads to the **same** final formula derived below — so the formula itself is completely general.

2 Areas of the Three Trapeziums

Do we know how to find the area of a trapezium? Yes:

$$\text{area of a trapezium} \;=\; \tfrac{1}{2} \times (\text{sum of parallel sides}) \times (\text{height})$$

Now we must quantify these for each trapezium.

The big trapezium $ADFC$

Its parallel sides are AD and CF, and its height is the perpendicular distance along the X axis, namely DF.

Since $A = (x_1, y_1)$ and dropping the perpendicular makes the y coordinate vanish, we get $D = (x_1, 0)$. So the length of AD is purely in terms of y:

$$AD \;=\; y_1 \qquad \text{and similarly} \qquad CF \;=\; y_3.$$

For the height, subtract the lowest from the highest along the axis: $F = (x_3, 0)$ and $D = (x_1, 0)$, so

$$DF \;=\; x_3 - x_1.$$

$$\text{ar}(ADFC) \;=\; \tfrac{1}{2}\,(y_1 + y_3)\,(x_3 - x_1)$$

The two smaller trapeziums

In the same manner, for $ADEB$ the parallel sides are y_1 and y_2, and the height is ED:

$$\text{ar}(ADEB) \;=\; \tfrac{1}{2}\,(y_1 + y_2)\,(x_2 - x_1) \qquad\qquad \text{ar}(BEFC) \;=\; \tfrac{1}{2}\,(y_2 + y_3)\,(x_3 - x_2)$$

$ADFC$	$ADEB$	$BEFC$
sides $y_1,\ y_3$	sides $y_1,\ y_2$	sides $y_2,\ y_3$
height $x_3 - x_1$	height $x_2 - x_1$	height $x_3 - x_2$

3 A Worked Example

Take the triangle plotted above, with

$$A \;=\; (4,\ 2), \qquad B \;=\; (6,\ 1), \qquad C \;=\; (8,\ 3).$$

Computing each piece

Here $D = (4, 0)$, so the length of AD is 2; likewise $CF = 3$, and $DF = 8 - 4 = 4$.

$$\text{ar}(ADFC) \;=\; \tfrac{1}{2}\,(2 + 3) \times 4 \;=\; 10$$

$$\text{ar}(ADEB) \;=\; \tfrac{1}{2}(2+1)\times 2 \;=\; 3$$
$$\text{ar}(BEFC) \;=\; \tfrac{1}{2}(1+3)\times 2 \;=\; 4$$

Therefore

$$\text{ar}(\triangle ABC) \;=\; 10-3-4 \;=\; \textbf{3} \text{ square units.}$$

4 The General Formula

Writing out the expression in general,

$$\text{ar}(\triangle ABC) \;=\; \tfrac{1}{2}(y_1+y_3)(x_3-x_1) \;-\; \tfrac{1}{2}(y_1+y_2)(x_2-x_1) \;-\; \tfrac{1}{2}(y_2+y_3)(x_3-x_2)$$

and juggling with it — expanding the cross products and simplifying — gives a very neat expression.

Area of a triangle from coordinates

$$\text{ar}(\triangle ABC) \;=\; \frac{1}{2}\Big|\, x_1(y_2-y_3) \;+\; x_2(y_3-y_1) \;+\; x_3(y_1-y_2) \,\Big|$$

Two points to note

The absolute value ensures that the area never comes out negative. With it, the vertices may be taken in **any order** — the formula always gives the correct area.
Without the absolute value, the expression inside gives a *signed* area: it is **positive** when the vertices are taken in the **anti-clockwise** direction and **negative** when they are taken clockwise.

Checking the example against the formula

With $A\,(4,2)$, $B\,(6,1)$, $C\,(8,3)$:

$$\text{ar} \;=\; \tfrac{1}{2}\big|\,4(1-3)+6(3-2)+8(2-1)\,\big| \;=\; \tfrac{1}{2}\big|\,-8+6+8\,\big| \;=\; \tfrac{1}{2}\times 6 \;=\; 3$$

which agrees with the trapezium calculation.

So far we have seen: given two points, how to find the **distance** between them; given three collinear points, how the **section formula** gives the ratio or the coordinates of the dividing point; and now, for non-collinear points, how to compute the **area** of the triangle using the coordinate system.

Lecture 5 — Slope of a Line

Rise over run, the inclination θ, and why $m = \tan\theta$ is consistent

What this lecture covers

After the area of a triangle, let us turn again to a two-point system, but now to a one-dimensional object — a **line**. We know from basic classes that two points uniquely determine a line. If we are given two points, we should be able to characterise the line passing through them. Relating that geometric object algebraically to coordinate geometry needs the concept of the **slope** of a line.

1 What Is Slope?

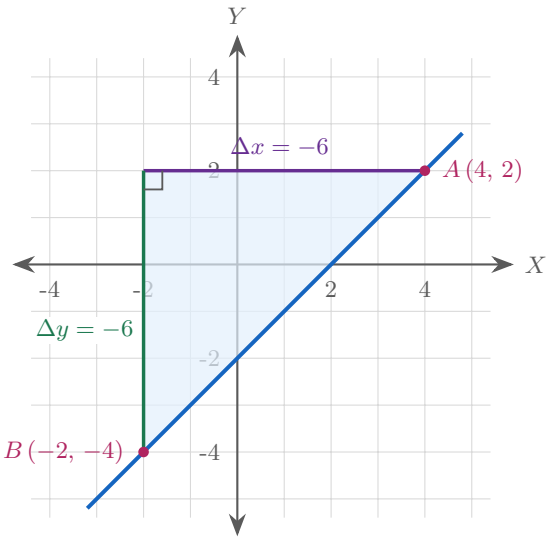
In a vague way, what do we understand by the slope of a line? Looking at the coordinate plane: if we move some units in the x direction, what is the corresponding change in the y direction?

To answer this we take a ratio — the change in the y direction to the change in the x direction. Some people call it the **rise over run** ratio: *run* is the horizontal direction, *rise* the vertical one.

$$\text{slope} = \frac{\text{rise}}{\text{run}} = \frac{\text{change in } y}{\text{change in } x} = \frac{\Delta y}{\Delta x}$$

An example

Take a line with two points marked on it. As usual, we construct a right-angled triangle using those two points. The vertical leg is the movement of the line in the y direction in going from one point to the other, and the horizontal leg is the movement in the x direction.



Reading off the slope

Going from $(4, 2)$ to $(-2, -4)$:

$$\Delta y = -4 - 2 = -6 \qquad \Delta x = -2 - 4 = -6$$

so the slope of the line is

$$m = \frac{-6}{-6} = 1.$$

2 The Formal Definition

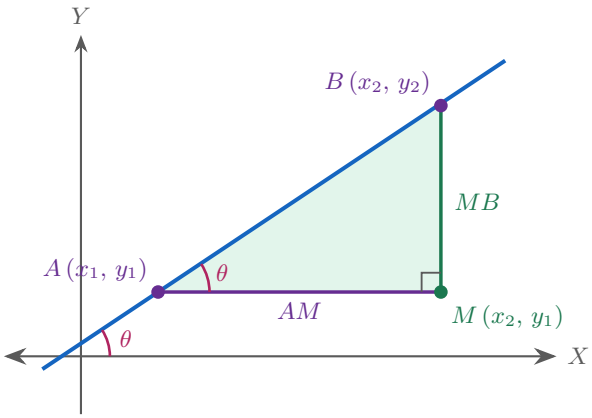
Given two points on a line in the coordinate plane, identify them as (x_1, y_1) and (x_2, y_2), and construct a right-angled triangle whose third vertex is

$$M = (x_2,\ y_1).$$

Once that is done we know the change in the x direction and the corresponding change in the y direction, so we can compute their ratio.

There is also an angle formed at the vertex A of this right-angled triangle; call it θ. Recalling trigonometry, this ratio is exactly a trigonometric ratio:

$$m = \frac{MB}{AM} = \frac{y_1 - y_2}{x_1 - x_2} = \tan\theta$$



Why the two angles are equal

The line AM is parallel to the X axis, and the given line intersects the X axis. So by the basics of geometry the angle at A and the angle the line makes with the X axis are the same angle θ.

Order of subtraction

It does not matter whether we take $y_1 - y_2$ or $y_2 - y_1$, as long as we do the same thing on top and bottom:

$$\frac{y_1 - y_2}{x_1 - x_2} = \frac{y_2 - y_1}{x_2 - x_1}$$

Since we are finally taking a ratio, the swap cancels — just keep the two in sync so that no confusion arises.

Two terms

- m is called the **slope** of the line.
- θ is called the **inclination** of the line with respect to the positive X axis, measured in the **anti-clockwise** direction.

3 How Far Can θ Go?

The inclination may be 0; it may be 90°, where tan is not defined; and it may go up to 180°. So the variation allowed is

$$0^\circ \leq \theta < 180^\circ$$

Salient features

Line parallel to the X axis

The angle of inclination is 0°, therefore

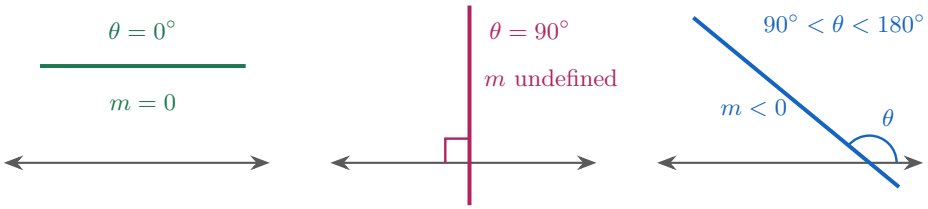
$$m = \tan 0^\circ = 0.$$

Line at 90° — along the Y axis

Here $\tan 90^\circ$ is undefined, therefore the **slope is undefined**.
With an inclination of 90° the line is the Y axis, or a line of the form

$$x = \text{constant}$$

and one can have any amount of movement in the y direction with *no* change in x — which itself creates the problem. So the slope is undefined at $\theta = 90^\circ$.



4 The Second Definition, and an Ambiguity

Slope via inclination

If θ is the inclination of a line ℓ, then $\tan\theta$ is called the **slope** or **gradient** of the line:

$$m = \tan\theta, \qquad \theta \neq 90^\circ.$$

This matches the original definition, but there may appear to be a glitch — an ambiguity — because θ is the angle made with the *positive* X axis. Let us resolve it.

Where the ambiguity shows up

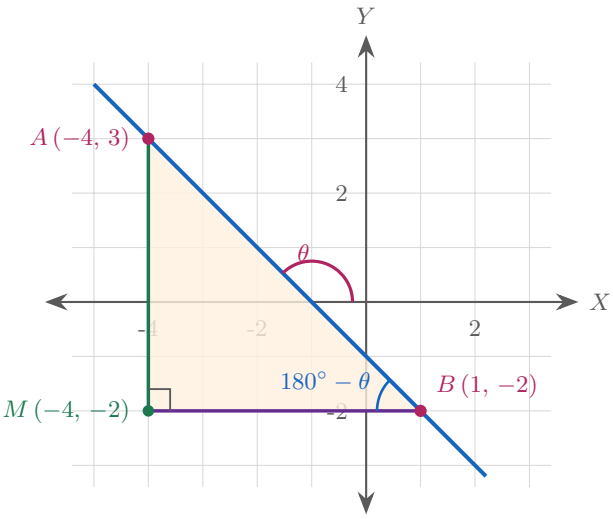
Consider a line whose inclination θ is an **obtuse** angle. How do we evaluate the tangent of such an angle, and will it contradict our first definition of slope?

Using the rise-over-run formula, we construct a right-angled triangle with the right angle at

$$M = (x_1, y_2)$$

— in the figure below, the point $(-4, -2)$. The angle of this triangle at B — the vertex where the horizontal leg MB meets the line — is

$$180^\circ - \theta.$$



Resolving it

The right-angled triangle only tells us the *size* of the rise and of the run. Since its angle at B is $180° - \theta$,

$$\frac{MA}{MB} \;=\; \tan(180° - \theta) \;=\; -\tan\theta$$

by a simple trigonometric identity — and this is a *positive* number, because θ is obtuse and so $\tan\theta < 0$.

But this line *falls* as we move to the right, so Δy and Δx have **opposite signs**, and the rise-over-run ratio is the *negative* of the ratio of lengths:

$$m \;=\; \frac{y_1 - y_2}{x_1 - x_2} \;=\; -\frac{MA}{MB} \;=\; -(-\tan\theta) \;=\; \tan\theta$$

Checking with the figure

With $A(-4, 3)$ and $B(1, -2)$:

$$m \;=\; \frac{3 - (-2)}{-4 - 1} \;=\; \frac{5}{-5} \;=\; -1$$

which is negative, as it must be for an obtuse inclination — here $\theta = 135°$, and $\tan 135° = -1$. So $m = \tan\theta$ exactly.

The conclusion

The formula for the slope is **consistent** no matter which definition we use: $m = \Delta y/\Delta x = \tan\theta$ for acute and obtuse inclinations alike. Therefore the slope of a line is **uniquely determined** once the line is given.

Lecture 6 — Using Slope: Parallel, Perpendicular and Intersecting Lines

Conditions on slopes, the angle between two lines, and collinearity

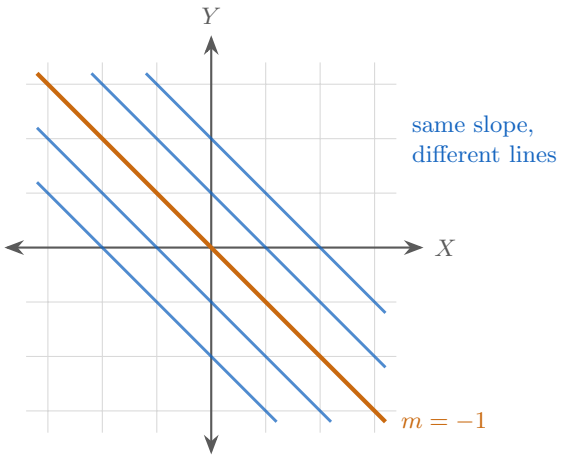
What this lecture covers

Given a line we can uniquely determine its slope — but given a slope, can we uniquely determine a line? Asking that question leads us to the real usefulness of slope: conditions for two lines to be **parallel**, for them to be **perpendicular**, a formula for the **angle** at which two lines intersect, and a test for three points to be **collinear**.

1 Does a Slope Determine a Line?

In other words: can there be many lines with the same slope?

Fix one line — say one whose slope works out to -1 from its coordinates. Any other line with the same **inclination** has the same tangent, and therefore the same slope. There are infinitely many such parallel lines.



The answer

No — a slope, or an inclination, does *not* uniquely determine a line. What stays constant across a whole family of parallel lines is the inclination.

This is exactly what makes the concept useful: if the inclinations are the same, the lines must be parallel. Rotating by 90° gives a condition for perpendicular lines, and considering two general intersecting lines gives a relation for their angle of intersection.

2 Condition for Parallel Lines

Let ℓ_1 and ℓ_2 be two **non-vertical** lines with slopes m_1, m_2 and inclinations α, β respectively.

Why non-vertical?

Vertical lines have an inclination of 90°, for which the concept of slope is undefined. So we need non-vertical lines throughout.

If the lines are parallel, the slopes are equal

$$\ell_1 \parallel \ell_2 \implies \alpha = \beta \implies \tan\alpha = \tan\beta \implies m_1 = m_2$$

since $\tan\alpha$ is the slope of ℓ_1 and $\tan\beta$ is the slope of ℓ_2.

The converse

Assume the slopes are equal. Then by definition $\tan\alpha = \tan\beta$. Does that imply $\alpha = \beta$?

Why the converse works

Because we restrict inclinations to lie between $0°$ and $180°$, the value of tan determines the angle uniquely on that range. So α and β lying in $[0°, 180°)$ forces

$$\alpha = \beta$$

which means the inclinations are the same, and hence $\ell_1 \parallel \ell_2$.

Characterisation of parallel lines

Two non-vertical lines ℓ_1 and ℓ_2 are parallel **if and only if**

$$m_1 = m_2$$

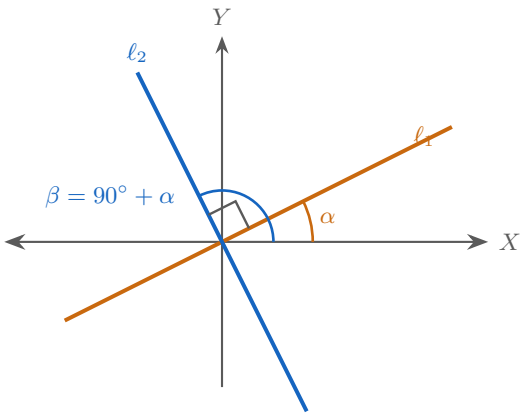
So to check whether two lines are parallel, it suffices to check whether their slopes are equal.

3 Condition for Perpendicular Lines

Now take two perpendicular non-vertical lines ℓ_1 and ℓ_2, with slopes m_1, m_2 and inclinations α, β.

If $\ell_1 \perp \ell_2$, then

$$\beta = 90° + \alpha.$$



Taking the tangent of β:

$$\tan\beta = \tan(90° + \alpha) = -\cot\alpha = -\frac{1}{\tan\alpha}$$

But $\tan\beta = m_2$ and $\tan\alpha = m_1$, so

$$m_2 = -\frac{1}{m_1} \qquad \Longleftrightarrow \qquad m_1\, m_2 = -1$$

The converse

So far we have shown only that perpendicularity forces the product of the slopes to be -1. Now suppose $m_1 m_2 = -1$. Then

$$\tan\alpha \cdot \tan\beta = -1 \qquad \Longrightarrow \qquad \tan\alpha = -\frac{1}{\tan\beta} \qquad \Longrightarrow \qquad \tan\alpha = -\cot\beta.$$

And $-\cot\beta$ is the tangent of either $\beta + 90^\circ$ or $\beta - 90^\circ$, since

$$\tan(\beta + 90^\circ) = -\cot\beta \qquad \text{and} \qquad \tan(\beta - 90^\circ) = -\cot\beta.$$

In either case the difference between α and β is 90°, and therefore $\ell_1 \perp \ell_2$.

Characterisation of perpendicular lines

Two non-vertical lines ℓ_1 and ℓ_2 are perpendicular **if and only if**

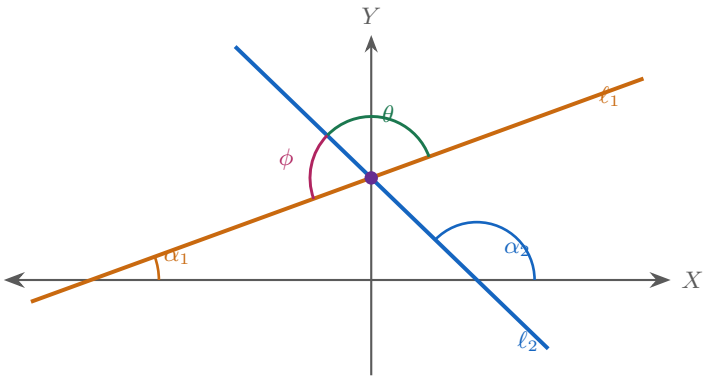
$$m_1\, m_2 = -1$$

that is, the product of their slopes equals -1.

4 The Angle Between Two Intersecting Lines

What if the lines are neither parallel nor perpendicular? If they are not parallel they must intersect. Given their slopes, can we talk about the angle of intersection? Yes.

Let ℓ_1 and ℓ_2 be non-vertical lines with slopes m_1, m_2 and inclinations α_1, α_2, intersecting at some point, and let θ and ϕ be the two **adjacent angles** formed, so that $\theta + \phi = 180^\circ$. (If they met perpendicularly both adjacent angles would be 90° — a case we have already resolved.)



From the figure, α_2 is obtuse and α_1 is acute, so

$$\theta = \alpha_2 - \alpha_1, \qquad \alpha_1,\ \alpha_2 \neq 90^\circ$$

the restriction being there because we cannot work with vertical lines.

Bringing in the slopes

Take the tangent of θ and use the standard trigonometric formula:

$$\tan\theta \;=\; \tan(\alpha_2 - \alpha_1) \;=\; \frac{\tan\alpha_2 - \tan\alpha_1}{1 + \tan\alpha_1 \tan\alpha_2}$$

But $\tan\alpha_2 = m_2$ and $\tan\alpha_1 = m_1$, so

Angle between two lines

$$\tan\theta \;=\; \frac{m_2 - m_1}{1 + m_1 m_2} \qquad\qquad \tan\phi \;=\; \frac{m_1 - m_2}{1 + m_1 m_2} \qquad (m_1 m_2 \neq -1)$$

The second follows because $\phi = 180^\circ - \theta$, and we have already seen that

$$\tan(180^\circ - \theta) \;=\; -\tan\theta,$$

so the numerator simply swaps to $m_1 - m_2$ while the denominator stays the same. The condition $m_1 m_2 \neq -1$ remains, since the lines must not be perpendicular.

5 Collinearity Using Slopes

One more interesting question: what if three points are collinear — how are the slopes interpreted?

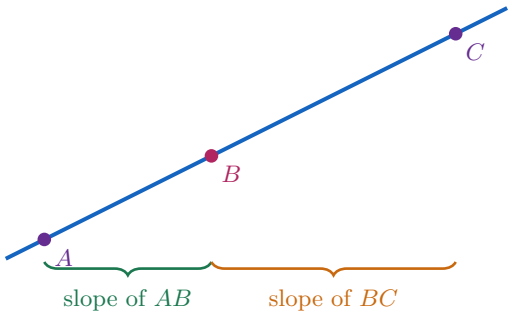
If three points lie on the same line, the slopes computed between them must all be equal, since they lie on one line and share a common point.

Collinearity condition

Three points A, B, C are collinear if and only if

$$\text{slope of } AB \;=\; \text{slope of } BC$$

where B is the common point between the two segments.



equal slopes and a shared point $\Rightarrow$ A, B, C collinear

So if there is a common point among the three, and the slopes are equal, the points are collinear — that is the relation of collinearity expressed through slopes.

Lecture 7 — Representation of a Line

Horizontal and vertical lines, the point–slope form and the two–point form

What this lecture covers

We have seen how slope relates to a line, and how it decides whether lines are parallel or perpendicular. But slope alone cannot represent a line uniquely. So what *is* required to represent a line uniquely — and, given any point, how do we decide whether it lies on the line? These two questions lead us to the equations of a line.

1 The Two Questions

What we are after

1. How do we represent a line **uniquely**?
2. Given any point, how do we decide whether it **lies on** the line?

Rephrasing the first: for a given line ℓ we want a **definite condition or expression** describing the line in terms of the coordinate plane — that is, what condition must the coordinates satisfy in order to describe ℓ?

Once we understand that condition, the second question is automatically answered. If the coordinates of a point P satisfy the condition for ℓ, then P lies on ℓ; otherwise it does not. It becomes a simple matter of checking.

2 Horizontal Lines

Horizontal line

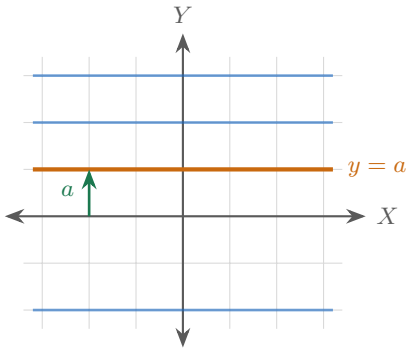
A line is a **horizontal line** if and only if it is **parallel to the X axis**.

There are infinitely many horizontal lines. To specify one uniquely, what do we need to know? The distance of the line from the X axis — that is, its y coordinate.

Let that value be a. Since the line is horizontal, all its points lie at the same distance from the X axis, so every point on it satisfies

$$y = a \qquad \text{with points of the form } (x,\ a)$$

Here x may take any value, but the y coordinate is fixed at a.



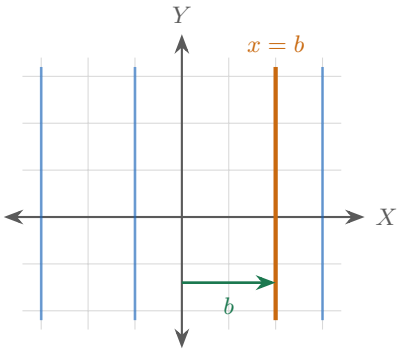
3 Vertical Lines

Vertical line

A line is a **vertical line** if and only if it is **parallel to the Y axis**.

To locate such a line we need its distance from the Y axis — that is, the value it takes on the x coordinate. Identify any one point on the line; if that distance is b, then all points of the form $(b,\ y)$ lie on the line, and

$$x = b \qquad \text{with points of the form } (b,\ y)$$



Recognising them from points

If two points have the same x coordinate and we know they lie on a line, then the line is parallel to the Y axis — a vertical line. Likewise, if two points share the same y coordinate, the line is parallel to the X axis — a horizontal line.

4 An Example with Both

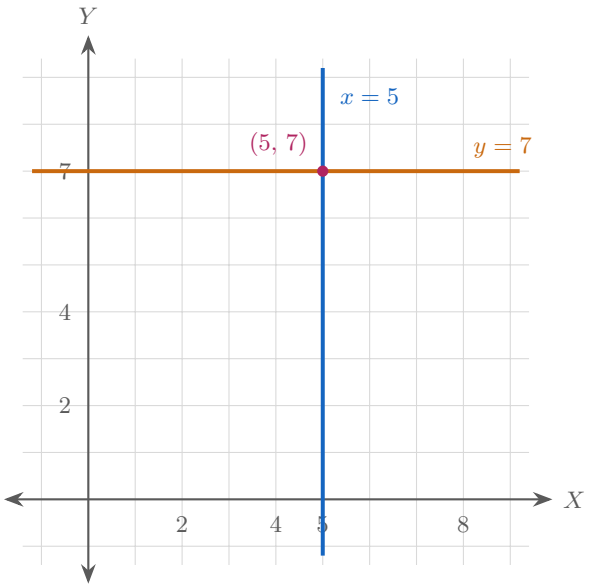
Lines parallel to the axes through $(5, 7)$

The lines are parallel to the axes and pass through $(5, 7)$. A line parallel to the X axis is horizontal; a line parallel to the Y axis is vertical. So the question asks for one of each.
Horizontal line. We need its distance from the X axis, which is 7, since $(5, 7)$ lies on it:

$$y = 7$$

Vertical line. We need its distance from the Y axis, which is 5; all points on it are of the form $(5,\ y)$, and $(5, 7)$ is one of them:

$$x = 5$$



Slopes of these two lines

A **vertical line** has inclination 90°, and therefore its slope is **undefined**.
A **horizontal line** does not cross the X axis (unless it *is* the X axis, $y = 0$), but its inclination with respect to the X axis is 0°, so its slope is **0**.

Having disposed of the cases where the slope is 0 or does not exist, we now need similar expressions for lines that are *not* vertical.

5 The Point–Slope Form

We know a slope alone cannot determine a line. But what if we are given a slope *together with* a point?

The problem

For a non-vertical line ℓ with slope m and a fixed point $P(x_0,\ y_0)$ on it, find the algebraic representation of the line.

We know that two points uniquely determine a line, so take another point $Q(x,\ y)$ whose coordinates are unknown but which is assumed to lie on ℓ. From the definition of slope as change in y over change in x, applied to P and Q:

$$m = \frac{y - y_0}{x - x_0}$$

No need for $\tan\theta$ here

This has nothing to do with $\tan\theta$ directly — although if the inclination were known we could find it. Since nothing specific is given about the angle, we work with the slope, which already encodes $\tan\theta$.

Multiplying throughout by $x - x_0$:

Point–slope form

$$y - y_0 \;=\; m\,(x - x_0)$$

This condition uniquely identifies the line — no other line satisfies it. So any point lying on the line must satisfy it.

What we have achieved

The geometric object is now represented by an equation. We began by representing a *point* — a geometric object — through its coordinates; now we represent an *infinite set of points* satisfying a condition, namely a line, algebraically.

A worked example

Line through $(5, 6)$ with slope -2

Take $Q\,(x, y)$ as an arbitrary point on the line. By the point–slope form:

$$-2 \;=\; \frac{y - 6}{x - 5}$$

Rearranging:

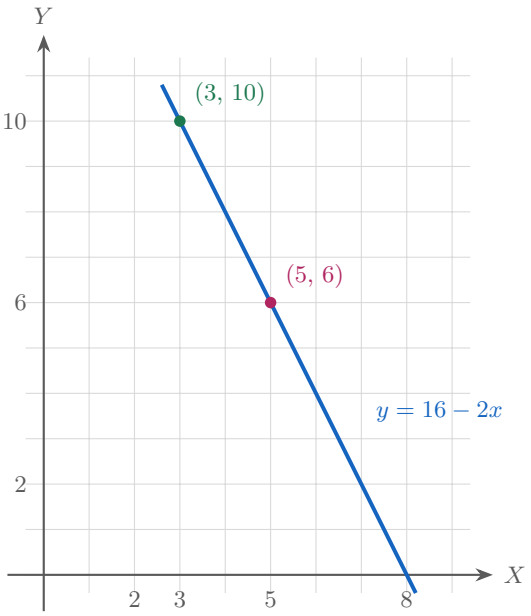
$$y - 6 \;=\; 2\,(5 - x) \qquad \Longrightarrow \qquad y \;=\; 16 - 2x$$

Checking with a second point

Which value of y satisfies the equation when $x = 3$?

$$y \;=\; 16 - 2(3) \;=\; 10$$

So the point $(3, 10)$ must lie on this line. Since two points uniquely determine a line, we can draw the line through $(5, 6)$ and $(3, 10)$.



Why draw the graph?

The question did not ask for a graph, but drawing one always verifies whether the answer is correct — so it is good practice to cross-check.

6 The Two–Point Form

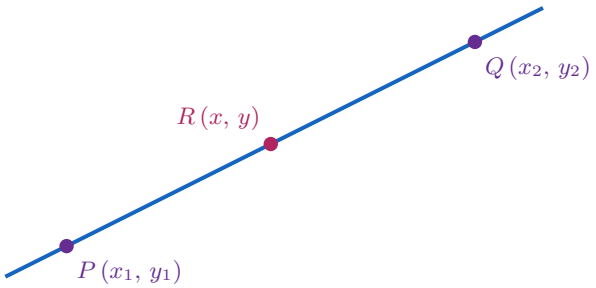
Suppose instead of a slope we are simply given **two points**. Can we find the equation of the line? Obviously yes — from two points we can always compute the slope, and then use the point–slope form. But we can also derive a form of its own.

The set-up

Let the line ℓ pass through $P\,(x_1, y_1)$ and $Q\,(x_2, y_2)$, and let $R\,(x, y)$ be an **arbitrary** point on ℓ.

Since P, Q and R all lie on one line, they are **collinear**. Therefore the slope computed from P and R must equal the slope computed from P and Q:

$$\text{slope of } PR \;=\; \text{slope of } PQ$$
$$\frac{y - y_1}{x - x_1} \;=\; \frac{y_2 - y_1}{x_2 - x_1}$$



P, Q, R collinear $\Rightarrow$ slope PR = slope PQ

Notice that the right-hand side is nothing but the slope m — this is very similar to the point–slope form, except that instead of computing the slope explicitly we keep it as a ratio. Taking the denominator to the other side:

Two–point form

$$y - y_1 \;=\; \frac{y_2 - y_1}{x_2 - x_1}\,(x - x_1)$$

Any point lying on the line must satisfy this condition, and the line is uniquely characterised by it.

A worked example

Line through $(5, 10)$ and $(-4, -2)$

Take $R\,(x, y)$ as an arbitrary point and apply the two–point form:

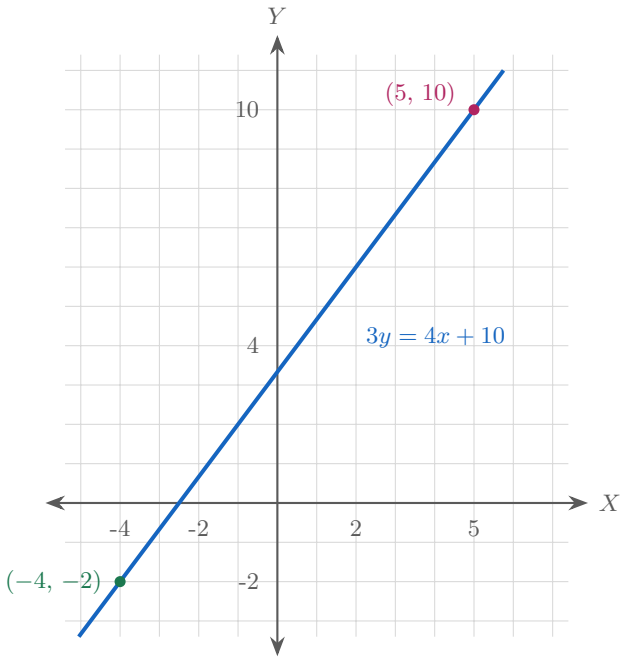
$$\frac{y - 10}{x - 5} \;=\; \frac{-2 - 10}{-4 - 5} \;=\; \frac{-12}{-9} \;=\; \frac{4}{3}$$

The order does not matter — starting from the other point gives

$$\frac{10-(-2)}{5-(-4)} = \frac{12}{9} = \frac{4}{3}$$

the same answer. So it makes no difference which point we call (x_1, y_1).
Simplifying:

$$3\,(y-10) = 4\,(x-5) \qquad \Longrightarrow \qquad 3y = 4x+10$$



Verifying both points

$$(5,10): \quad 3(10)=30, \quad 4(5)+10=30 \qquad \checkmark$$

$$(-4,-2): \quad 3(-2)=-6, \quad 4(-4)+10=-6 \qquad \checkmark$$

Lecture 8 — Intercept Forms of a Line

Slope–intercept form, the x-intercept version, and the intercept form

What this lecture covers

The two primary forms we have are the **two–point form** and the **point–slope form**. Let us now look at some special variations in which the calculations become extremely easy. Taking the point–slope form with a special point gives the **slope–intercept form**, and taking two special points gives the elegant **intercept form**.

1 The y-Intercept and the Slope–Intercept Form

y-intercept

Let ℓ be a line with slope m that cuts the Y axis at the point c. Then c is called the **y-intercept** of ℓ.

What does it mean to cut the Y axis at c? The y coordinate of that point is c and its x coordinate is 0. So the point of intersection with the Y axis is

$$(0,\ c) \qquad \text{and it lies on } \ell.$$

Now apply the point–slope form, using this special point in place of a general (x_0, y_0):

$$y - c \ = \ m\,(x - 0) \qquad \Longrightarrow \qquad y - c \ = \ mx$$

Slope–intercept form

$$\boxed{y \ = \ mx + c}$$

the standard form we generally deal with for straight lines.

Why it is convenient

If we know the y-intercept c and the slope m, there is **no calculation at all** — we simply write down $y = mx + c$. If the intercept is not available, we fall back on the point–slope form and work it out.

2 The x-Intercept Version

A line cutting the Y axis may equally cut the X axis.

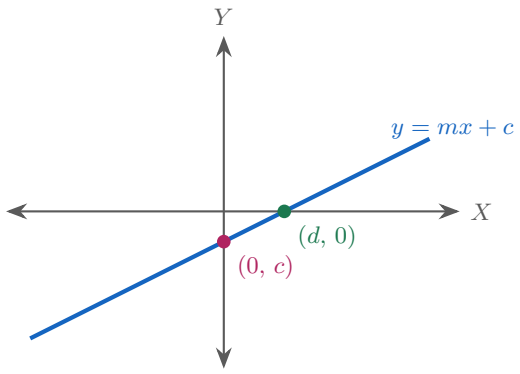
x-intercept

If a line ℓ with slope m cuts the X axis at the point d, then d is called the **x-intercept** of ℓ. The point of intersection is

$$(d,\ 0) \qquad \text{and it lies on } \ell.$$

Using the point–slope form again with this point:

$$y - 0 \;=\; m\,(x - d) \qquad \Longrightarrow \qquad y \;=\; mx - md$$



3 Worked Examples

Example 1 — slope $\frac{1}{2}$, y-intercept $-\frac{3}{2}$

Here things are very easy, since we need only $y = mx + c$ with $m = \frac{1}{2}$ and $c = -\frac{3}{2}$:

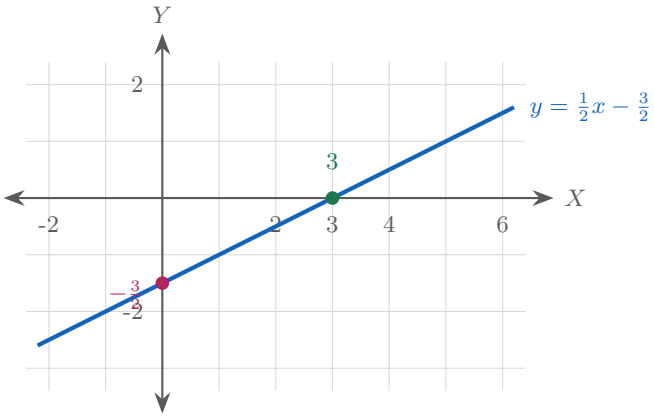
$$y \;=\; \tfrac{1}{2}x \;-\; \tfrac{3}{2}$$

Finding its x-intercept

Setting $y = 0$:

$$0 \;=\; \tfrac{1}{2}x - \tfrac{3}{2} \qquad \Longrightarrow \qquad x \;=\; 3$$

So the x-intercept of this line is 3. The question could equally have been asked as "find the equation of a line with slope $\frac{1}{2}$ and x-intercept 3" — and the answer would be the same line.



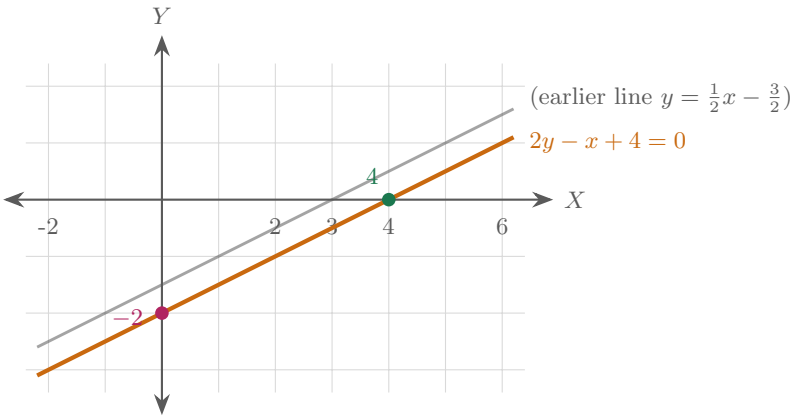
Example 2 — slope $\frac{1}{2}$, x-intercept 4

This is definitely *not* the same line, since the x-intercept is 4 rather than 3 — but the slope is still $\frac{1}{2}$. Since equal slopes mean the lines are **parallel**, this line must be parallel to the previous one, cutting the X axis at a different point.
Using $y = mx - md$ with $d = 4$ and $m = \frac{1}{2}$:

$$y \;=\; \tfrac{1}{2}\,(x - 4)$$

Simplifying:

$$2y \;=\; x - 4 \qquad \Longrightarrow \qquad 2y - x + 4 \;=\; 0$$



4 The Intercept Form

Suppose instead we are given *both* intercepts — how do we identify the line?

The set-up

Let a line make an x-intercept at a and a y-intercept at b. Then it passes through the two points

$$(a,\ 0) \qquad \text{and} \qquad (0,\ b).$$

We use the **two–point form**, taking $(a, 0)$ as the first point:

$$y - 0 \ = \ \frac{b-0}{0-a}\,(x-a)$$

Dividing throughout by b:

$$\frac{y}{b} \ = \ -\frac{x}{a} + 1$$

and shifting the negative term to the left-hand side:

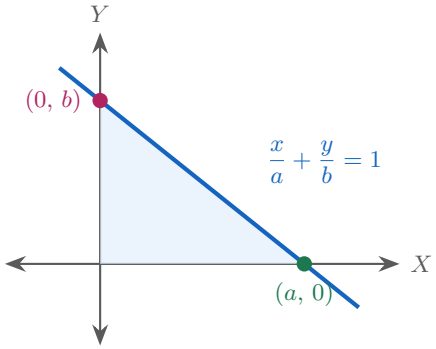
Intercept form

$$\boxed{\frac{x}{a} + \frac{y}{b} \ = \ 1}$$

Nothing to memorise

See how neat this is: the x-intercept is a, so a goes below x; the y-intercept is b, so b goes below y. It is simply

$$\frac{x}{x\text{-intercept}} + \frac{y}{y\text{-intercept}} \ = \ 1.$$



A worked example

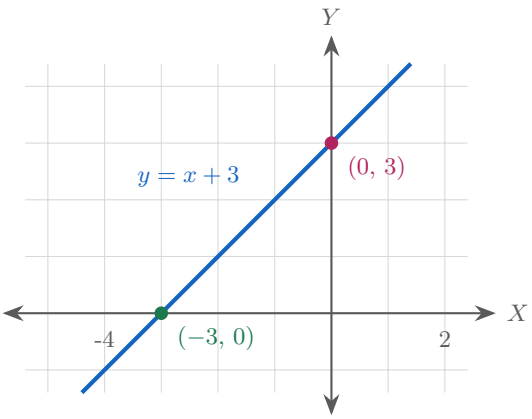
Line with x-intercept -3 and y-intercept 3

No complicated calculation is needed:

$$\frac{x}{-3} + \frac{y}{3} = 1$$

Multiplying throughout by 3:

$$-x + y = 3 \qquad \Longrightarrow \qquad y = x + 3$$



It is always better to verify using a graph: the line passing through $(-3, 0)$ and $(0, 3)$ is indeed $y = x + 3$.

Lecture 9 — The General Equation of a Line

$Ax + By + C = 0$, its correspondence with the earlier forms, and reading geometry off an equation

What this lecture covers

So far we have represented a line — a geometric object — algebraically, through various forms of equations. It is time to recollect those forms, notice what they have in **common**, and write down a single general equation of a line that will be useful for further analysis.

1 The Forms We Have Studied

We had two primary forms, each with a specialised version:

Form	Given	Equation
Point–slope	point (x_0, y_0), slope m	$y - y_0 = m\,(x - x_0)$
Slope–intercept (y)	y-intercept c, slope m	$y = mx + c$
Slope–intercept (x)	x-intercept d, slope m	$y = m\,(x - d)$
Two–point	(x_1, y_1), (x_2, y_2)	$y - y_1 = \dfrac{y_2 - y_1}{x_2 - x_1}(x - x_1)$
Intercept	x-intercept a, y-intercept b	$\dfrac{x}{a} + \dfrac{y}{b} = 1$

How they are related

The slope–intercept forms are specialisations of the point–slope form: putting $(x_0, y_0) = (d, 0)$ gives $y - 0 = m(x - d)$, and putting $(x_0, y_0) = (0, c)$ gives $y = mx + c$.
The two–point form is itself closely related to the point–slope form: given two points we can always determine the slope, so m is simply replaced by the ratio of the change in y to the change in x, and (x_0, y_0) by (x_1, y_1).
The intercept form comes from the two–point form with $(x_1, y_1) = (a, 0)$ and $(x_2, y_2) = (0, b)$:

$$y \;=\; \frac{b}{-a}\,(x - a) \qquad \Longrightarrow \qquad \frac{x}{a} + \frac{y}{b} \;=\; 1$$

— below x write the x-intercept, below y write the y-intercept, and equate to 1.

2 The Common Feature

Look at all these forms. In the point–slope form, x_0 and y_0 are fixed and the slope m is fixed; what we are identifying is a **condition on the coordinates**, so the **variables** are x and y.

The same feature is visible in every form: an expression of the shape

$$(\text{constant}) \times y \;+\; (\text{constant}) \times x \;+\; (\text{constant}).$$

Separating variables from constants

Take the point–slope form and rewrite it:

$$y - y_0 \;=\; m\,(x - x_0) \qquad \Longrightarrow \qquad y - mx \;=\; y_0 - mx_0$$

Here $y_0 - mx_0$ is the constant of the equation, the variable y carries the coefficient 1, and the variable x carries the coefficient $-m$.

Similarly, in the intercept form $\frac{x}{a} + \frac{y}{b} = 1$, the coefficient attached to x is $\frac{1}{a}$, the coefficient attached to y is $\frac{1}{b}$, and there is a constant term.

General form

$$\boxed{Ax \;+\; By \;+\; C \;=\; 0}$$

3 Matching the General Form with the Earlier Ones

Rewriting the point–slope form so as to compare:

$$-mx + y - \big(y_0 - mx_0\big) \;=\; 0$$

(we may of course multiply throughout by any constant, which changes nothing). Comparing with $Ax + By + C = 0$:

$$m \;=\; -\frac{A}{B} \qquad\qquad y_0 - mx_0 \;=\; -\frac{C}{B}$$

Reading off each form

Slope–intercept (y). Putting $y_0 = c$ and $x_0 = 0$, the term mx_0 becomes irrelevant, so

$$c \;=\; -\frac{C}{B}$$

Slope–intercept (x). Putting $x_0 = d$ and $y_0 = 0$,

$$d \;=\; -\frac{C}{A}$$

Intercept form. The same exercise gives

$$a \;=\; -\frac{C}{A} \qquad\qquad b \;=\; -\frac{C}{B}$$

Two–point form. Here m is replaced by the ratio of the two differences, and (x_0, y_0) by (x_1, y_1) — the slope is still $-A/B$ and the constant still $-C/B$.

slope	**x-intercept**	**y-intercept**
$m \;=\; -\frac{A}{B}$	$a \;=\; d \;=\; -\frac{C}{A}$	$b \;=\; c \;=\; -\frac{C}{B}$

So there is an exact one-to-one correspondence between the general equation and each of the earlier forms.

4 Why the General Equation Is Better

When we derived the earlier forms, we always assumed **non-vertical** lines, because for vertical lines the slope does not exist. Those lines are of the form $x =$ constant.

Vertical lines are covered

Putting $B = 0$ in the general equation:

$$Ax + C = 0 \qquad \Longrightarrow \qquad x = -\frac{C}{A}$$

which is exactly a vertical line — and agrees with what the intercept correspondence reveals. Horizontal lines are covered too: putting $A = 0$ gives

$$y = -\frac{C}{B}$$

a line parallel to the X axis (equivalently, $m = 0$).

So both vertical and non-vertical lines are handled, which is why this is the **general** form of the equation of a line.

Linear equations in two variables

In earlier classes you may have met $Ax + By + C$ (without the "$= 0$") as a **polynomial in two variables** — and it is a *linear* polynomial in two variables. Hence the term **linear equation in two variables**.

One necessary condition

For this to represent a line, A and B **cannot be simultaneously zero**. If both were zero we would be equating a constant to 0, which is invalid.
Individually they may be zero: $A = 0$ gives $y =$ constant, a line parallel to the X axis, and $B = 0$ gives $x =$ constant, a line parallel to the Y axis.

General linear equation

Any equation of the form

$$Ax + By + C = 0, \qquad A,\ B \text{ not simultaneously } 0,$$

is called a **general linear equation**, or the general equation of a line.

What this identification buys us

A linear polynomial in two variables, equated to zero, gives a line. This identifies the geometric object — the straight line — with an algebraic representation. It works both ways: we need not discuss the line at all, but may discuss its algebraic representation instead; or we may start from an algebraic representation and deduce the geometric properties of the line.

5 A Worked Example

The problem

The equation of a line is

$$3x - 4y + 12 = 0.$$

Find its slope, its x-intercept and its y-intercept.

Comparing with $Ax + By + C = 0$:

$$A = 3, \qquad B = -4, \qquad C = 12.$$

The intercepts

$$x\text{-intercept} = -\frac{C}{A} = -\frac{12}{3} = -4$$

$$y\text{-intercept} = -\frac{C}{B} = -\frac{12}{-4} = 3$$

The slope

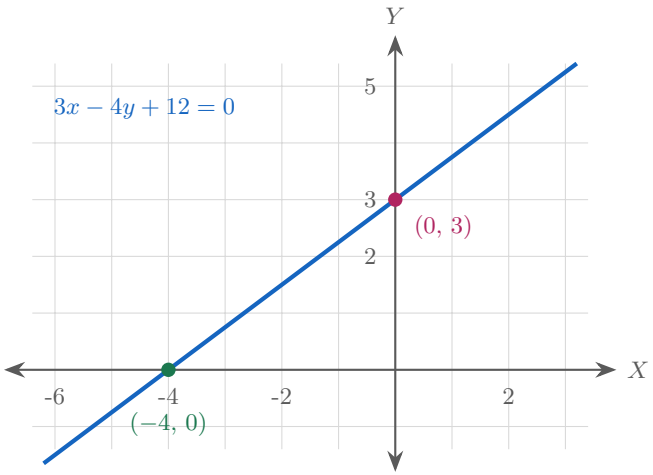
Rearranging into the form $y = mx + c$ — push the $4y$ to the right-hand side and divide:

$$3x + 12 = 4y \qquad \Longrightarrow \qquad y = \frac{3}{4}x + 3$$

so the slope is

$$m = \frac{3}{4}$$

which agrees with $m = -A/B = -\dfrac{3}{-4} = \dfrac{3}{4}$.



Verifying graphically

Although the calculation may be correct, it is always better to verify. The slope is $\frac{3}{4}$, the x-intercept is -4 and the y-intercept is 3, and the drawn line passes through both intercepts. For further verification, pick any point on the line, put its coordinates into the equation, and check that the value comes out to 0.

A spot check

Take the point $(4, 6)$, which lies on the line:

$$3(4) - 4(6) + 12 \;=\; 12 - 24 + 12 \;=\; 0 \qquad \checkmark$$

Lecture 10 — Parallel and Perpendicular Lines in General Form

Conditions $a_1b_2 = a_2b_1$ and $a_1a_2 + b_1b_2 = 0$

What this lecture covers

Here is another application of the general form of the equation of a line. We take two lines written in general form and re-derive the conditions for them to be **parallel** and **perpendicular** — conditions we already know in terms of slope — so that they can be checked directly from the coefficients.

1 The Statement

The problem

Given two lines

$$a_1x + b_1y + c_1 = 0 \qquad a_2x + b_2y + c_2 = 0$$

with $b_1 \neq 0$ and $b_2 \neq 0$, show that they are

parallel if $a_1b_2 = a_2b_1$ and **perpendicular** if $a_1a_2 + b_1b_2 = 0$.

What $b_1, b_2 \neq 0$ means

It means the lines are **non-vertical**. Recall that putting $B = 0$ in $Ax + By + C = 0$ gives $x = -C/A$, a vertical line, for which the slope does not exist. You can verify this for yourself.

2 The Plan

How do we characterise parallel and perpendicular lines? We already have both answers in terms of slope:

$$\text{parallel} \iff m_1 = m_2 \qquad \text{perpendicular} \iff m_1m_2 = -1$$

So the job reduces to finding the slopes of the two lines from their general forms.

3 Finding the Slopes

Take the first line. We identify it with the slope–intercept form $y = mx + c$. Since $b_1 \neq 0$, we may divide throughout by b_1 and move the remaining terms to the right-hand side:

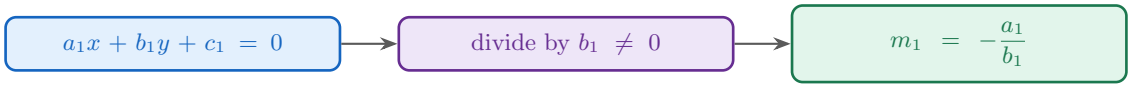
$$a_1x + b_1y + c_1 = 0 \quad \Longrightarrow \quad y = -\frac{a_1}{b_1}x - \frac{c_1}{b_1}$$

so the slope is

$$m_1 = -\frac{a_1}{b_1}.$$

Applying the same trick to the second line:

$$m_2 = -\frac{a_2}{b_2}$$



4 The Parallel Condition

Since $b_1, b_2 \neq 0$ we are dealing with non-vertical lines, and two non-vertical lines are parallel if and only if their slopes are equal. So put $m_1 = m_2$:

$$-\frac{a_1}{b_1} = -\frac{a_2}{b_2} \quad \Longrightarrow \quad \frac{a_1}{b_1} = \frac{a_2}{b_2}$$

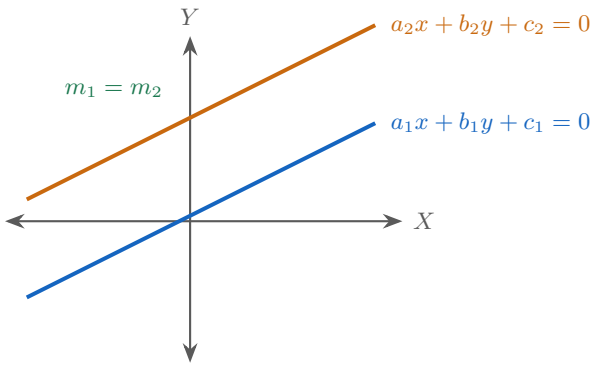
the minus signs cancelling each other. Now multiply both sides by b_1b_2, which is legal since neither is zero:

Condition for parallel lines

$$\boxed{a_1b_2 = a_2b_1}$$

Parallel or coincident?

Equal slopes also hold when the two equations describe the **same** line (for example $x-y+1=0$ and $2x-2y+2=0$). So $a_1b_2 = a_2b_1$ means the lines are **parallel or coincident**. They are *distinct* parallel lines when, in addition, the constant terms are not in the same proportion, i.e. $b_1c_2 \neq b_2c_1$.



5 The Perpendicular Condition

Similarly, for perpendicular lines the product of the slopes is -1. Multiply m_1 and m_2 and set the product equal to -1 — again the minus signs cancel:

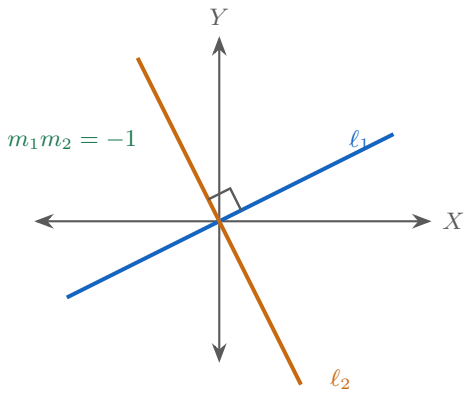
$$\left(\frac{a_1}{b_1}\right)\left(\frac{a_2}{b_2}\right) = -1$$

Taking the denominator b_1b_2 to the right-hand side:

$$a_1a_2 = -b_1b_2$$

Condition for perpendicular lines

$$\boxed{a_1a_2 \ + \ b_1b_2 \ = \ 0}$$



6 What We Have Achieved

A new characterisation

We have related the earlier characterisation of parallel and perpendicular lines — stated in terms of slope — to the **general form** of the equation. Given two non-vertical lines in general form, we need only check the coefficients:

$$\text{parallel:} \quad a_1b_2 = a_2b_1 \qquad\qquad \text{perpendicular:} \quad a_1a_2 + b_1b_2 = 0$$

No slopes need to be computed explicitly.

Relation	In terms of slope	In general form
Parallel (or coincident)	$m_1 = m_2$	$a_1b_2 = a_2b_1$
Perpendicular	$m_1m_2 = -1$	$a_1a_2 + b_1b_2 = 0$

A quick check of the conditions

Take $3x - 4y + 12 = 0$, so $a_1 = 3$, $b_1 = -4$.
Parallel test with $6x - 8y + 5 = 0$ ($a_2 = 6$, $b_2 = -8$):

$$a_1b_2 \ = \ 3 \times (-8) \ = \ -24, \qquad a_2b_1 \ = \ 6 \times (-4) \ = \ -24 \qquad \checkmark$$

Perpendicular test with $4x + 3y - 1 = 0$ ($a_2 = 4$, $b_2 = 3$):

$$a_1a_2 + b_1b_2 \ = \ 3 \times 4 \ + \ (-4) \times 3 \ = \ 12 - 12 \ = \ 0 \qquad \checkmark$$

Lecture 11 — A Worked Example on Perpendicular Lines

Finding the line perpendicular to a given line through a given point

What this lecture covers

We now put the general form and the perpendicularity condition to work on a concrete problem: given a line and a point not on it, find the line through that point perpendicular to the given one — and then verify the answer.

1 The Problem

The problem

Find the equation of a line **perpendicular** to

$$x - 2y + 3 = 0$$

and passing through the point $(-1,\ 2)$.

Step 1 — the slope of the given line

Identify the given equation with the general form $Ax + By + C = 0$:

$$A = 1, \qquad B = -2, \qquad C = 3.$$

So the slope of the given line is

$$m_1 = -\frac{A}{B} = -\frac{1}{-2} = \frac{1}{2}$$

Step 2 — the slope of the perpendicular line

If a line is perpendicular to it, the product of the slopes is -1:

$$m_1\, m_2 = -1 \qquad \Longrightarrow \qquad m_2 = -\frac{1}{m_1} = -\frac{1}{\frac{1}{2}} = -2$$

Step 3 — use the point–slope form

The problem now reduces to this: a line with slope -2 passing through $(-1, 2)$. Using

$$y - y_0 = m\,(x - x_0)$$

with $x_0 = -1$ and $y_0 = 2$:

$$y - 2 = -2\,\big(x - (-1)\big) = -2\,(x + 1)$$

Expanding, the constant 2 cancels:

$$y - 2 = -2x - 2 \qquad \Longrightarrow \qquad y = -2x$$

The answer

$$y = -2x \qquad \text{or, in general form,} \qquad 2x + y = 0$$

Verification

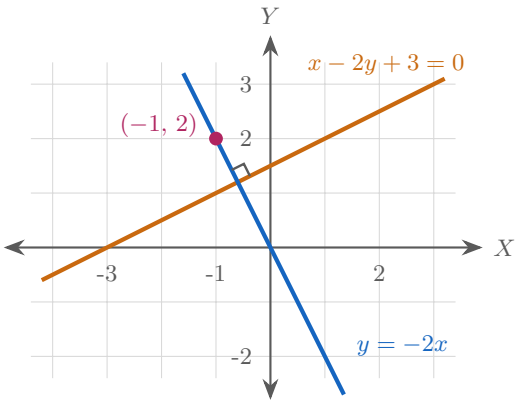
Two quick checks

Does the point lie on it?

$$2(-1) + 2 = -2 + 2 = 0 \qquad \checkmark$$

Are the lines perpendicular? Using the coefficient condition with $(a_1, b_1) = (1, -2)$ and $(a_2, b_2) = (2, 1)$:

$$a_1a_2 + b_1b_2 = (1)(2) + (-2)(1) = 2 - 2 = 0 \qquad \checkmark$$



Reading the graph

The orange line is the one given, $x - 2y + 3 = 0$; the point $(-1, 2)$ is marked, and the blue line through it is the one we found. The two lines meet at $(-0.6,\ 1.2)$ at an angle of $90°$, so the lines are perpendicular and the answer is correct — the verification test has passed.

Lecture 12 — Distance of a Point from a Line

The perpendicular distance formula, and the distance between two parallel lines

What this lecture covers

Having found a line perpendicular to a given line through a given point, a natural question follows: *what is the distance of that point from the line?* We derive the perpendicular distance formula using the area of a triangle, and then use it to find the distance between two parallel lines.

1 Setting Up the Problem

The goal

Find the distance of the point $P(x_1,\ y_1)$ from the line ℓ whose equation is given in general form:

$$Ax + By + C \ = \ 0$$

To understand the location of P relative to the line, we must first understand the line itself in terms of its geometric features — its slope, its intercepts, or points lying on it. Let us identify those.

Assuming $A \neq 0$, $B \neq 0$ and $C \neq 0$ (so that the line crosses both axes at two *distinct* points), we may rewrite the equation in **intercept form**:

$$\frac{x}{a} + \frac{y}{b} \ = \ 1 \qquad \text{with} \qquad a \ = \ -\frac{C}{A}, \qquad b \ = \ -\frac{C}{B}$$

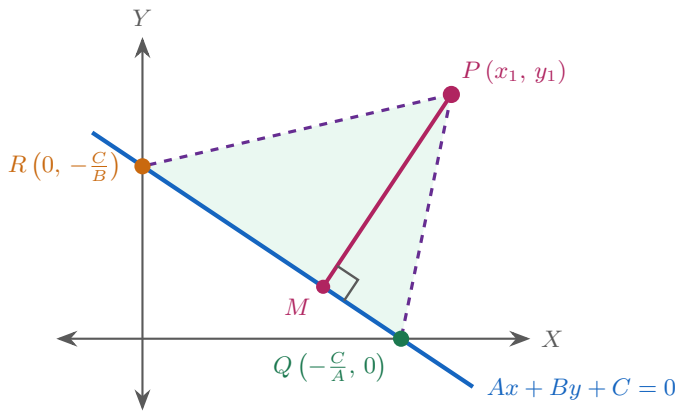
So the line passes through the two points

$$Q \ = \ \left(-\frac{C}{A},\ 0\right) \qquad\qquad R \ = \ \left(0,\ -\frac{C}{B}\right)$$

and these two points uniquely determine it, so we now know exactly how the line is located.

Which distance do we mean?

The distance must be the **shortest** distance from the point to the line — and the shortest one is the **perpendicular** distance. So the whole question reduces to finding the perpendicular segment PM.



2 Using the Triangle PQR

The dotted lines show the geometric object we can attach to this distance: the triangle PQR. Take QR as the base and PM as the height:

$$\text{ar}(\triangle PQR) \;=\; \tfrac{1}{2} \times \text{base} \times \text{height} \;=\; \tfrac{1}{2} \times QR \times PM$$

We do not know PM — but we already know how to compute the **area** of a triangle from the coordinates of its vertices, and we know how to compute the **length** QR. So rearranging:

The strategy

$$PM \;=\; \frac{2 \times \text{ar}(\triangle PQR)}{QR}$$

Step 1 — the area of $\triangle PQR$

Recall the area formula (because of the absolute value, the order of the vertices does not matter):

$$\text{ar} \;=\; \tfrac{1}{2} \left| x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2) \right|$$

For definiteness, start at P, then go to R, then to Q:

$$(x_1, y_1) = (x_1,\ y_1), \qquad (x_2, y_2) = \left(0,\ -\tfrac{C}{B}\right), \qquad (x_3, y_3) = \left(-\tfrac{C}{A},\ 0\right)$$

Substituting — the middle term vanishes because $x_2 = 0$:

$$\text{ar} \;=\; \tfrac{1}{2} \left| x_1 \left(-\frac{C}{B} - 0\right) \;+\; 0 \;+\; \left(-\frac{C}{A}\right)\left(y_1 + \frac{C}{B}\right) \right|$$

$$= \tfrac{1}{2} \left| -\frac{Cx_1}{B} - \frac{Cy_1}{A} - \frac{C^2}{AB} \right|$$

Taking C common inside the modulus and putting everything over the common denominator AB:

$$\text{ar}(\triangle PQR) \;=\; \frac{1}{2} \cdot \frac{|C|}{|AB|} \left| Ax_1 + By_1 + C \right|$$

Notice that the last factor is exactly the left-hand side of the general form, evaluated at P.

Step 2 — the length QR

This is easy: Q has only an x coordinate and R has only a y coordinate, so QR is the hypotenuse of a right-angled triangle whose legs lie along the axes:

$$QR \;=\; \sqrt{\frac{C^2}{A^2} + \frac{C^2}{B^2}} \;=\; \frac{|C|\sqrt{A^2 + B^2}}{|AB|}$$

taking C out and A, B out as $|AB|$.

Step 3 — putting them together

Feeding both into $PM = 2\,\text{ar}/QR$: the $\frac{1}{2}$ cancels with the 2, and the factors $|C|$ and $|AB|$ are the same on top and bottom, so they vanish too:

Distance of a point from a line

$$d = \frac{|Ax_1 + By_1 + C|}{\sqrt{A^2 + B^2}}$$

The special cases

The derivation assumed A, B, C all non-zero, so that Q and R exist and are distinct. The final formula, however, holds for **every** line. For instance, for a vertical line $x = k$ (i.e. $A = 1$, $B = 0$, $C = -k$) it gives $|x_1 - k|$, which is clearly the correct horizontal distance.

$$\text{ar} = \frac{1}{2} \cdot \frac{|C|}{|AB|} |Ax_1+By_1+C| \rightarrow QR = \frac{|C|\sqrt{A^2+B^2}}{|AB|} \rightarrow PM = \frac{|Ax_1+By_1+C|}{\sqrt{A^2+B^2}}$$

3 Distance Between Two Parallel Lines

This idea helps with one more thing: the distance between two **parallel** lines.

Since the lines are parallel they have the same slope m, so using the slope–intercept form:

$$\ell_1 : \; y = mx + c_1 \qquad \ell_2 : \; y = mx + c_2$$

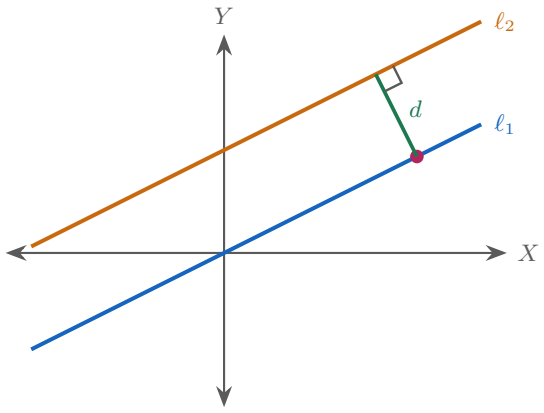
To use the previous formula we need a point. Take the x-intercept of ℓ_1 (this needs $m \neq 0$; for horizontal lines the distance is simply $|c_1 - c_2|$, which the final formula also gives). Writing ℓ_1 in general form,

$$A = -m, \qquad B = 1, \qquad C = -c_1$$

so its x-intercept is $-\dfrac{C}{A} = -\dfrac{c_1}{m}$, giving the point

$$\left(-\frac{c_1}{m},\ 0\right).$$

For ℓ_2, likewise, $A = -m$, $B = 1$, $C = -c_2$.



Substituting into the distance formula

The problem now reduces to the distance of the point $\left(-\frac{c_1}{m},\ 0\right)$ from the line ℓ_2. Putting $x_1 = -\frac{c_1}{m}$ and $y_1 = 0$ into $d = |Ax_1 + By_1 + C|/\sqrt{A^2 + B^2}$:

$$d \;=\; \frac{\left|\,(-m)\left(-\frac{c_1}{m}\right)+(1)(0)-c_2\,\right|}{\sqrt{(-m)^2+1^2}} \;=\; \frac{|\,c_1-c_2\,|}{\sqrt{1+m^2}}$$

The same thing in general form

We can also express this in the general form. Matching the two slope–intercept equations with general equations, we have

$$m \;=\; -\frac{A}{B}, \qquad c_1 \;=\; -\frac{C_1}{B}, \qquad c_2 \;=\; -\frac{C_2}{B}$$

so that the two lines become

$$Ax+By+C_1 \;=\; 0 \qquad\qquad Ax+By+C_2 \;=\; 0.$$

Substituting these identifications,

$$|c_1-c_2| \;=\; \frac{|C_1-C_2|}{|B|} \qquad \text{and} \qquad \sqrt{1+m^2} \;=\; \sqrt{1+\frac{A^2}{B^2}} \;=\; \frac{\sqrt{A^2+B^2}}{|B|},$$

and the common factor $|B|$ cancels between numerator and denominator:

Distance between two parallel lines

$$\boxed{d \;=\; \frac{|\,C_1-C_2\,|}{\sqrt{A^2+B^2}}}$$

This shows clearly the interconnection between the slope–intercept form and the general form, and gives the distance between two parallel lines from the point-to-line formula.

Make the coefficients match first

This formula needs both lines written with the **same** A and B. For example, $3x-4y+7=0$ and $6x-8y+5=0$ are parallel, but the second must first be divided by 2 (giving $3x-4y+\frac{5}{2}=0$) before using $C_1=7$, $C_2=\frac{5}{2}$.

4 Worked Examples

Example 1 — distance of a point from a line

Find the distance of the point $(3,\ -5)$ from the line

$$3x-4y-26 \;=\; 0.$$

Here $A=3$, $B=-4$, $C=-26$, and $(x_1,y_1)=(3,-5)$. Applying the formula:

$$d \;=\; \frac{|\,3(3)+(-4)(-5)+(-26)\,|}{\sqrt{3^2+(-4)^2}} \;=\; \frac{|\,9+20-26\,|}{\sqrt{9+16}} \;=\; \frac{3}{\sqrt{25}} \;=\; \frac{3}{5}$$

Example 2 — distance between two parallel lines

Find the distance between

$$3x-4y+7 \;=\; 0 \qquad \text{and} \qquad 3x-4y+5 \;=\; 0.$$

Here $C_1 = 7$ and $C_2 = 5$, with $A = 3$, $B = -4$. Applying the formula:

$$d = \frac{\left| 7 - 5 \right|}{\sqrt{3^2 + (-4)^2}} = \frac{2}{5}$$

Lecture 13 — Straight Line Fit

Vertical distance, sum of squared errors and regression

What this lecture covers

So far we have seen the distance of a point from a line, and the distance between two parallel lines. But is the perpendicular distance the only distance worth seeking? Here the **paradigm changes**: instead of one point we compare a whole *set* of points against a line, and we shall see that a different notion of distance is the right one.

1 A Physics Experiment

Consider an experiment you may have performed in the laboratory. Ohm's law states

$$V = IR$$

— voltage equals current times resistance, with voltage in volts, current in amperes and resistance in ohms. The task set by the physics teacher is to verify this law, or rather to use it to compute the resistance of a piece of equipment.

Relating it to the equation of a line

Replace the voltage V by y, the current I delivered to the circuit by x, and the unknown resistance R by m. What is the constant term? It is 0:

$$y = mx \qquad \text{where } y = \text{voltage}, \quad x = \text{current}, \quad m = \text{resistance}$$

and the whole purpose is to determine m.

Two pieces of prior information

The lab technician tells us that the lab has only **two** kinds of resistors: one of $1\,\Omega$ and one of $2\,\Omega$.
Also note that the line passes through the **origin** — if there is no current there is no voltage, so $(0, 0)$ is one point on it.

Since the line passes through $(0, 0)$, it seems that a *single* further reading should settle the matter: one point plus the origin determines the line. Let us see how that assumption works out.

2 The Observed Data

Various currents were passed and the corresponding voltages recorded:

Current x_i (A)	1	5	7	8	9	10
Voltage y_i (V)	2	4	8	9	8.7	9

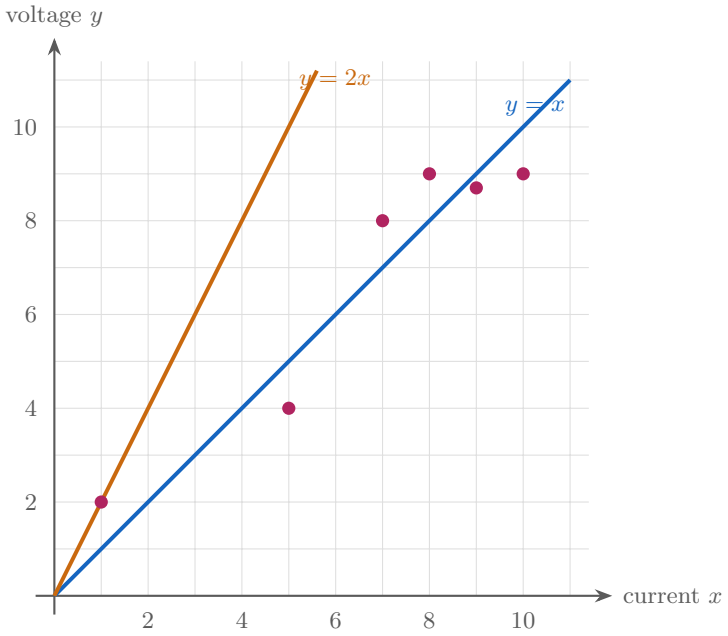
Stopping after one observation

If we take only the first reading $(1, 2)$ together with $(0, 0)$, the slope–intercept form with $c = 0$ gives

$$y = 2x$$

But with the same resistor we also obtained the other readings — and with more experimentation the data seem to follow a line much closer to

$$y = x.$$



So which line is the better fit? Note that $y = 2x$ passes through the point $(1, 2)$, while $y = x$ passes through none of the observed points exactly. The question must be settled mathematically.

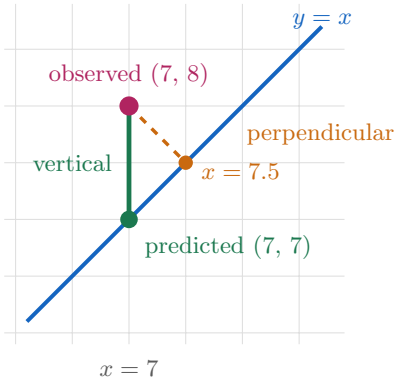
3 The Paradigm Shift

Let us zoom in and reconsider the **perpendicular distance**. Dropping a perpendicular from an observed point to the line gives a geometrically correct distance — but is it the right one *in this context*?

Why the perpendicular distance misleads

Suppose we pass a current of 7 amperes. The line $y = x$ says we should get 7 volts, but we actually recorded 8 volts.
If we drop a perpendicular from $(7, 8)$ to the line $y = x$, it meets the line at $x = 7.5$. But we are not interested in the value of y at 7.5 — we passed a current of 7 amperes, not 7.5 amperes.

So the perspective of distance must change: we want the discrepancy *at that particular value of* x. Instead of the perpendicular distance we therefore consider the **vertical distance** — the distance parallel to the Y axis.



The shift

For a set of observed points we measure the distance **parallel to the Y axis**, not perpendicular to the line. This is the paradigm shift: the perpendicular distance will not fetch us a useful result in the real-world setting.

4 Comparing the Two Lines

Both candidate lines, $y = x$ and $y = 2x$, pass through the origin — zero current, zero voltage — so the hypothesis is respected by both. Given the observations (x_i, y_i), which line is better?

Why we square the differences

If we take the line $y = x$, then feeding in x_i gives a predicted value x_i, while the actual output is y_i, so the discrepancy is $y_i - x_i$. For $y = 2x$ it is $y_i - 2x_i$.

But merely summing the differences is no good: some are positive and some negative, so they may cancel each other. We therefore **square** them.

Sum squared difference

$$\sum_{i=1}^{6}(y_i - x_i)^2 \qquad \sum_{i=1}^{6}(y_i - 2x_i)^2$$

Each term measures the **error** made — whether by the equipment or in the recording — between the observed value and the value the line predicts.

Working out both sums

x_i	y_i	$y_i - x_i$	$(y_i - x_i)^2$	$(y_i - 2x_i)^2$
1	2	1	1	0
5	4	−1	1	36
7	8	1	1	36
8	9	1	1	49
9	8.7	−0.3	0.09	86.49
10	9	−1	1	121
		Total	**5.09**	**328.49**

The conclusion

The line with the **least** sum squared difference is the better one:

$$5.09 \ll 328.49$$

so $y = x$ is a better line than $y = 2x$ — which is also intuitively evident from the figure, since the chunk of points is clustered around $y = x$. Therefore the resistance of the given equipment must be $\mathbf{1\,\Omega}$.

5 The General Notion

In this experiment we were lucky: the lab technician gave us only two possible resistance values. Real life is not so generous — usually no candidate values are supplied, and we must find the *best* line passing near a set of points. For that, this notion of the distance of a **set of points** from a line helps.

Sum squared error (SSE)

Given n points $(x_i,\ y_i)$, $i = 1$ to n, and a line

$$y \ = \ mx + c$$

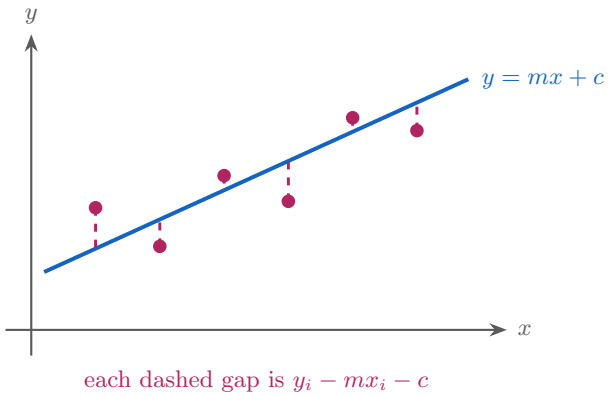
define

$$\boxed{\text{SSE} \ = \ \sum_{i=1}^{n} \big(y_i - mx_i - c\big)^2}$$

called the **sum squared error**, or sum squared distance.

A restriction

This equation is valid only when the line is **not vertical**. If it were vertical the form $y = mx + c$ would not apply — and in that case such an elaborate procedure would not be needed to estimate it anyway.



The question this raises

When handling a general problem we do not know what m and c should be. So the goal becomes:

The best-fit problem

Given a set of points, find the line $y = mx + c$ that **best fits** them — noting that the line need not pass through any of the points at all.
This splits into two questions: what values of m and c do we take, and what does "best fit" mean? Our answer to the second is **minimising the sum squared error**.

Why we stop here for now

We want the values of m and c that minimise SSE. This is beyond our present scope, since so far we have handled only *linear* terms, while these appear as *squares*. We shall need some strategies for carrying out such a minimisation, and these come in later videos of the course.

References and Acknowledgement

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Notes created by: Vishal (Instructor)